ARNAV CHITIKENA -- 2520030331

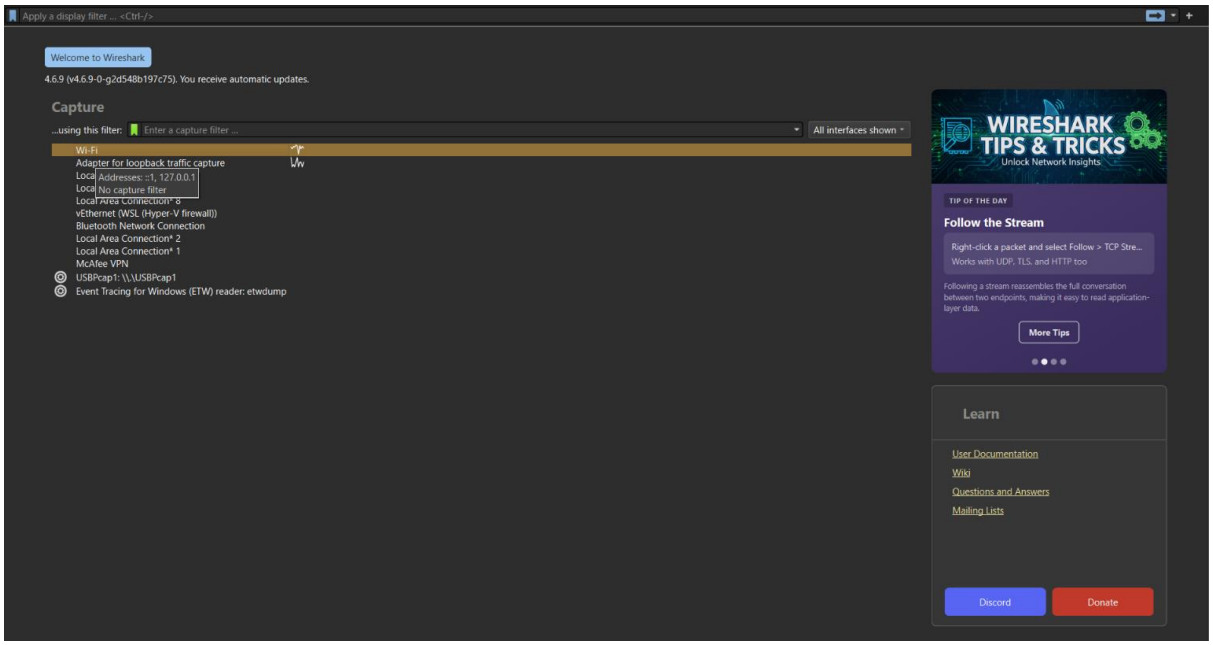
Experiment – 1

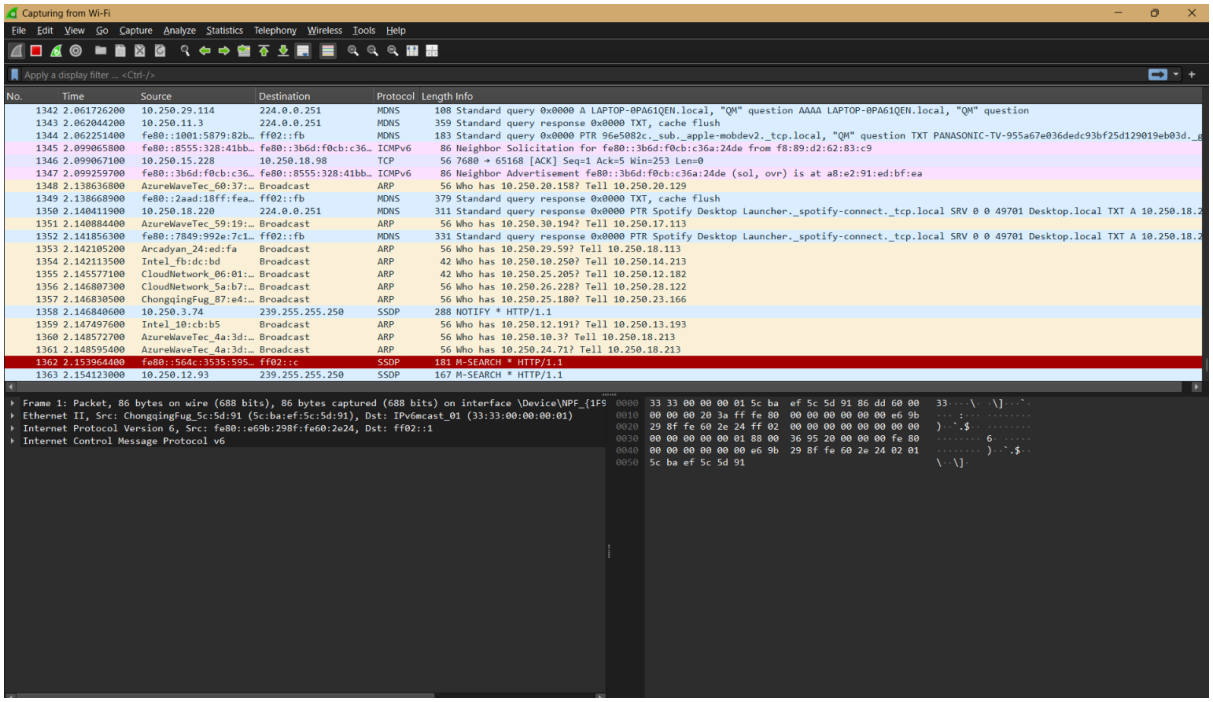
Experiment Name

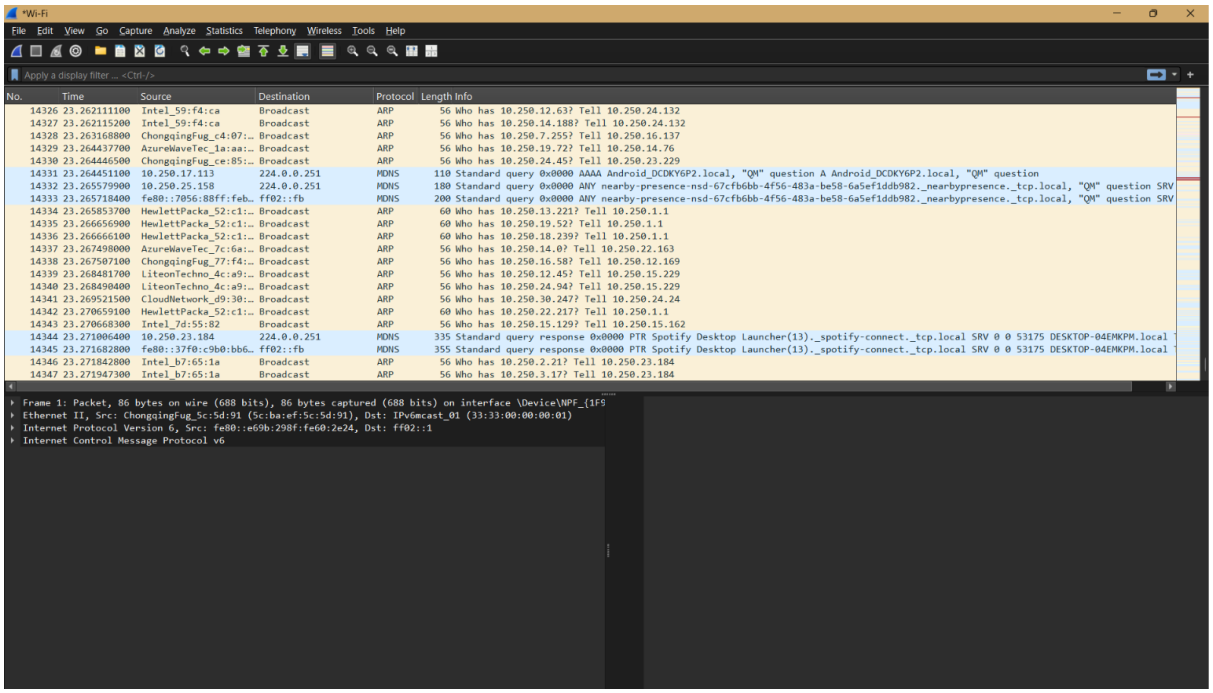
Encapsulation Builder: Analyse the Overhead Introduced by Protocol Headers and Determine How Encapsulation Affects Network Efficiency.

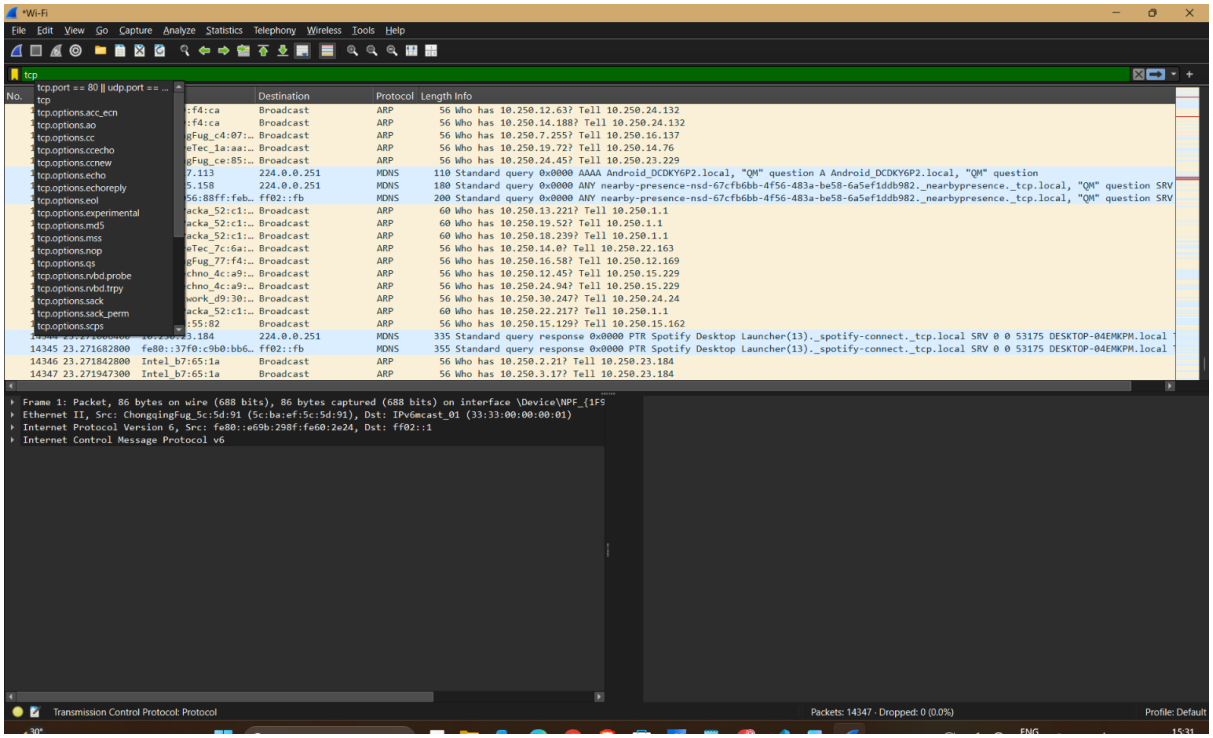
1. Aim

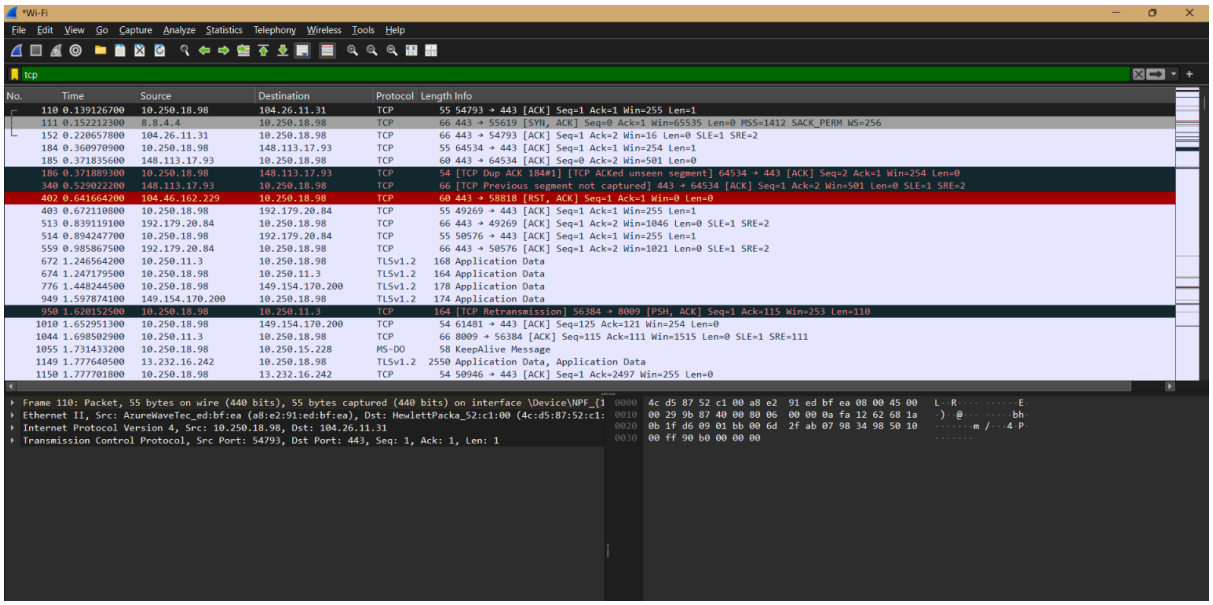
To capture network packets using Wireshark and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. To calculate the total overhead and determine network efficiency.

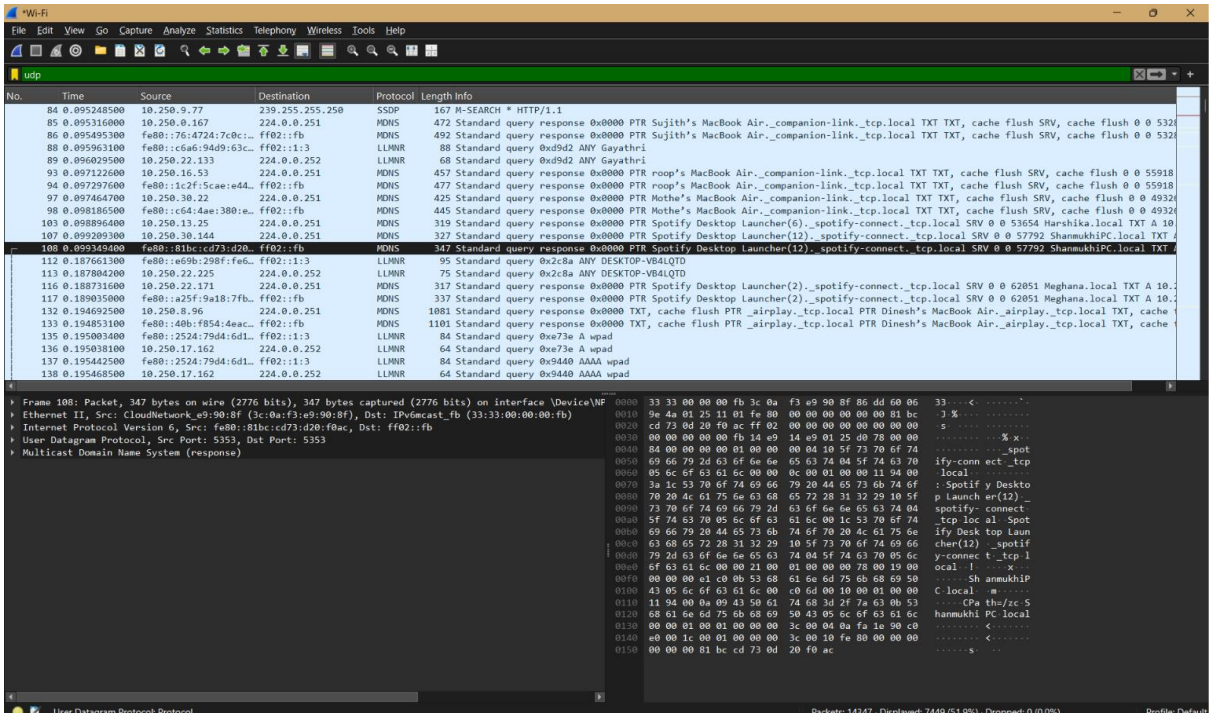


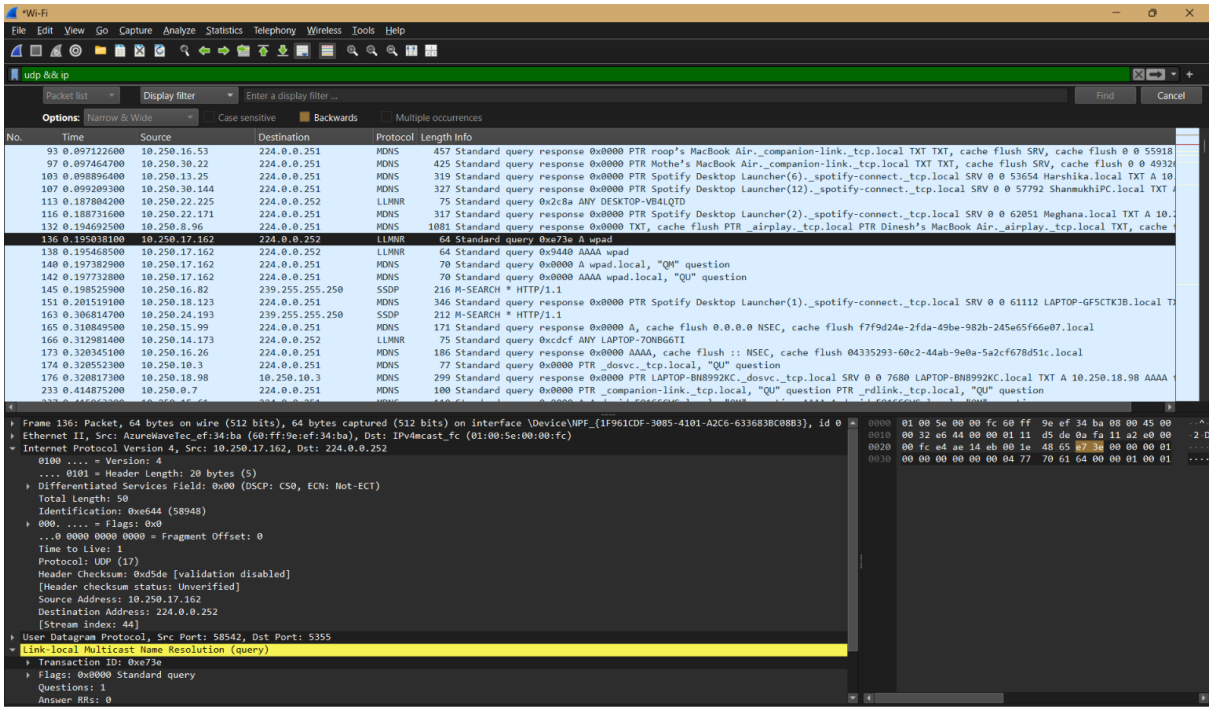

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1342 2.061726200 10.250.29.114 224.0.0.251 MDNS 108 Standard query 0x0000 A LAPTOP-0PA61QEN.local, "QM" question AAAA LAPTOP-0PA61QEN.local, "QM" question
1343 2.062044200 10.250.11.3 224.0.0.251 MDNS 359 Standard query response 0x0000 TXT, cache flush
1344 2.062251400 fe80::1001:5879:82b... ff02::fb MDNS 183 Standard query 0x0000 PTR 96e5082c._sub._apple-mobdev2._tcp.local, "QM" question TXT PANASONIC-TV-955a67e036dedc93bf25d129019eb03d._
1345 2.099065800 fe80::8555:328:41bb... fe80::3b6d:f0cb:c36... ICMPv6 86 Neighbor Solicitation for fe80::3b6d:f0cb:c36a:24de from f8:89:d2:62:83:c9
1346 2.099067100 10.250.15.228 10.250.18.98 TCP 56 7680 → 65168 [ACK] Seq=1 Ack=5 Win=253 Len=0
1347 2.099259700 fe80::3b6d:f0cb:c36... fe80::8555:328:41bb... ICMPv6 86 Neighbor Advertisement fe80::3b6d:f0cb:c36a:24de (sol, ovr) is at a8:e2:91:ed:bf:ea
1348 2.138636800 AzureWaveTec_60:37:... Broadcast ARP 56 Who has 10.250.20.158? Tell 10.250.20.129
1349 2.138668900 fe80::2aad:18ff:fea... ff02::fb MDNS 379 Standard query response 0x0000 TXT, cache flush
1350 2.140411900 10.250.18.220 224.0.0.251 MDNS 311 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 49701 Desktop.local TXT A 10.250.18.2
1351 2.140884400 AzureWaveTec_59:19:... Broadcast ARP 56 Who has 10.250.30.194? Tell 10.250.17.113
1352 2.141856300 fe80::7849:992e:7c1... ff02::fb MDNS 331 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 49701 Desktop.local TXT A 10.250.18.2
1353 2.142105200 Arcadyan_24:ed:fa Broadcast ARP 56 Who has 10.250.29.59? Tell 10.250.18.113
1354 2.142113500 Intel_fb:dc:bd Broadcast ARP 42 Who has 10.250.10.250? Tell 10.250.14.213
1355 2.145577100 CloudNetwork_06:01:... Broadcast ARP 42 Who has 10.250.25.205? Tell 10.250.12.182
1356 2.146807300 CloudNetwork_5a:b7:... Broadcast ARP 56 Who has 10.250.26.228? Tell 10.250.28.122
1357 2.146830500 ChongqingFug_87:e4:... Broadcast ARP 56 Who has 10.250.25.180? Tell 10.250.23.166
1358 2.146840600 10.250.3.74 239.255.255.250 SSDP 288 NOTIFY * HTTP/1.1
1359 2.147497600 Intel_10:cb:b5 Broadcast ARP 56 Who has 10.250.12.191? Tell 10.250.13.193
1360 2.148572700 AzureWaveTec_4a:3d:... Broadcast ARP 56 Who has 10.250.10.3? Tell 10.250.18.213
1361 2.148595400 AzureWaveTec_4a:3d:... Broadcast ARP 56 Who has 10.250.24.71? Tell 10.250.18.213
1362 2.153964400 fe80::564c:3535:595... ff02::c SSDP 181 M-SEARCH * HTTP/1.1
1363 2.154123000 10.250.12.93 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
Frame 1: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{1F9
Ethernet II, Src: ChongqingFug_5c:5d:91 (5c:ba:ef:5c:5d:91), Dst: IPv6mcast_01 (33:33:00:00:00:01)
Internet Protocol Version 6, Src: fe80::e69b:298f:fe60:2e24, Dst: ff02::1
Internet Control Message Protocol v6
0000 33 33 00 00 00 01 5c ba ef 5c 5d 91 86 dd 60 00
0010 00 00 00 20 3a ff fe 80 00 00 00 00 00 00 e6 9b
0020 29 8f fe 60 2e 24 ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 01 88 00 36 95 20 00 00 00 fe 80
0040 00 00 00 00 00 00 e6 9b 29 8f fe 60 2e 24 02 01
0050 5c ba ef 5c 5d 91

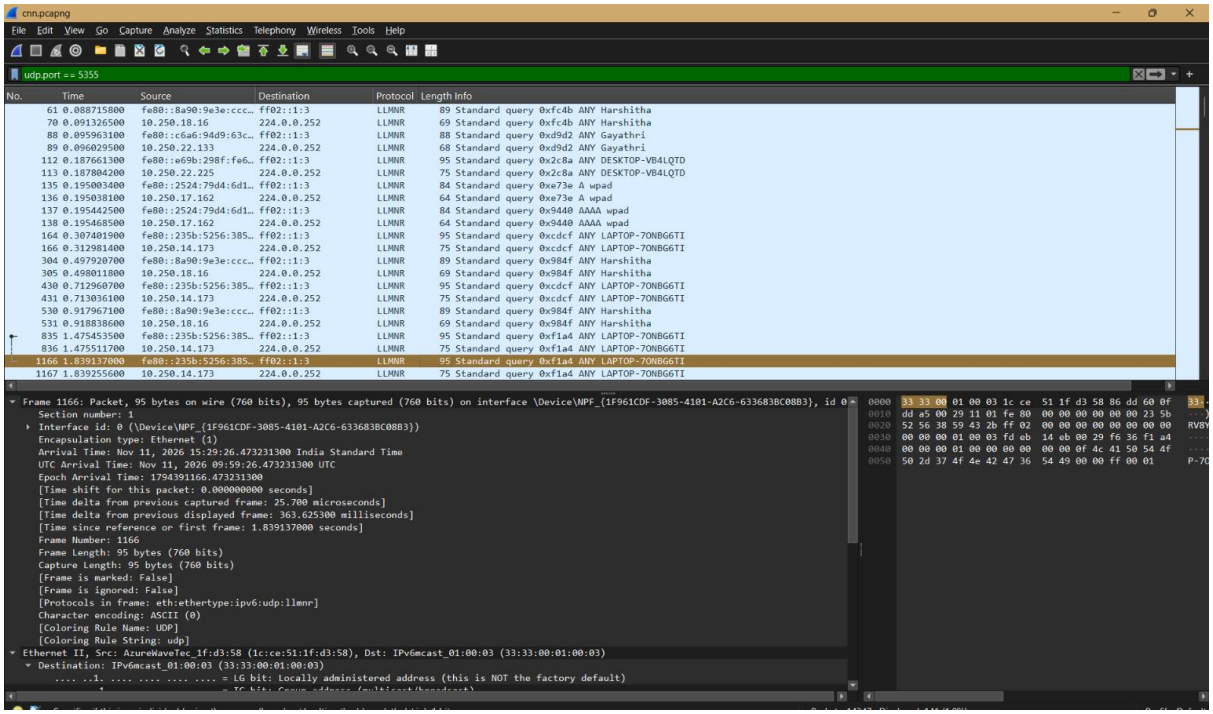

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
14326 23.262111100 Intel_59:f4:ca Broadcast ARP 56 Who has 10.250.12.63? Tell 10.250.24.132
14327 23.262115200 Intel_59:f4:ca Broadcast ARP 56 Who has 10.250.14.188? Tell 10.250.24.132
14328 23.263168800 ChongqingFug_c4:07:... Broadcast ARP 56 Who has 10.250.7.255? Tell 10.250.16.137
14329 23.264437700 AzureWaveTec_1a:aa:... Broadcast ARP 56 Who has 10.250.19.72? Tell 10.250.14.76
14330 23.264446500 ChongqingFug_ce:85:... Broadcast ARP 56 Who has 10.250.24.45? Tell 10.250.23.229
14331 23.264451100 10.250.17.113 224.0.0.251 MDNS 110 Standard query 0x0000 AAAA Android_DCDKY6P2.local, "QM" question A Android_DCDKY6P2.local, "QM" question
14332 23.265579900 10.250.25.158 224.0.0.251 MDNS 180 Standard query 0x0000 ANY nearby-presence-nsd-67cfb6bb-4f56-483a-be58-6a5ef1ddb982._nearbypresence._tcp.local, "QM" question SRV
14333 23.265718400 fe80::7056:88ff:feb... ff02::fb MDNS 200 Standard query 0x0000 ANY nearby-presence-nsd-67cfb6bb-4f56-483a-be58-6a5ef1ddb982._nearbypresence._tcp.local, "QM" question SRV
14334 23.265853700 HewlettPacka_52:c1:... Broadcast ARP 60 Who has 10.250.13.221? Tell 10.250.1.1
14335 23.266656900 HewlettPacka_52:c1:... Broadcast ARP 60 Who has 10.250.19.52? Tell 10.250.1.1
14336 23.266666100 HewlettPacka_52:c1:... Broadcast ARP 60 Who has 10.250.18.239? Tell 10.250.1.1
14337 23.267498000 AzureWaveTec_7c:6a:... Broadcast ARP 56 Who has 10.250.14.0? Tell 10.250.22.163
14338 23.267507100 ChongqingFug_77:f4:... Broadcast ARP 56 Who has 10.250.16.58? Tell 10.250.12.169
14339 23.268481700 LiteonTechno_4c:a9:... Broadcast ARP 56 Who has 10.250.12.45? Tell 10.250.15.229
14340 23.268490400 LiteonTechno_4c:a9:... Broadcast ARP 56 Who has 10.250.24.94? Tell 10.250.15.229
14341 23.269521500 CloudNetwork_d9:30:... Broadcast ARP 56 Who has 10.250.30.247? Tell 10.250.24.24
14342 23.270659100 HewlettPacka_52:c1:... Broadcast ARP 60 Who has 10.250.22.217? Tell 10.250.1.1
14343 23.270668300 Intel_7d:55:82 Broadcast ARP 56 Who has 10.250.15.129? Tell 10.250.15.162
14344 23.271006400 10.250.23.184 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(13)._spotify-connect._tcp.local SRV 0 0 53175 DESKTOP-04EMKPM.local
14345 23.271682800 fe80::37f0:c9b0:bb6... ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Launcher(13)._spotify-connect._tcp.local SRV 0 0 53175 DESKTOP-04EMKPM.local
14346 23.271842800 Intel_b7:65:1a Broadcast ARP 56 Who has 10.250.2.21? Tell 10.250.23.184
14347 23.271947300 Intel_b7:65:1a Broadcast ARP 56 Who has 10.250.3.17? Tell 10.250.23.184
Frame 1: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{1F9
Ethernet II, Src: ChongqingFug_5c:5d:91 (5c:ba:ef:5c:5d:91), Dst: IPv6mcast_01 (33:33:00:00:00:01)
Internet Protocol Version 6, Src: fe80::e69b:298f:fe60:2e24, Dst: ff02::1
Internet Control Message Protocol v6

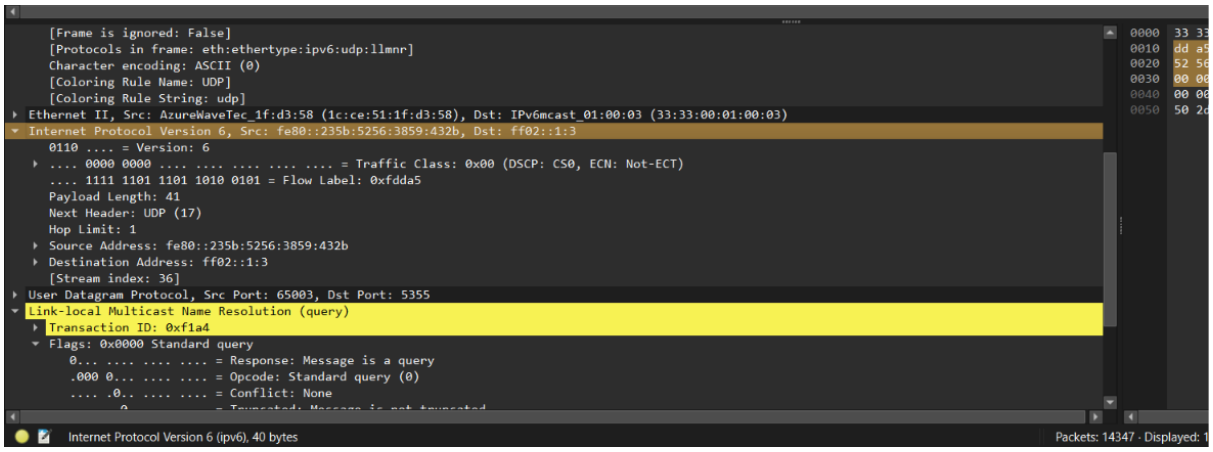

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
tcp.port == 80 || udp.port == ...
tcp
tcp.options.acc_ecn
tcp.options.ao
tcp.options.cc
tcp.options.ccecho
tcp.options.ccnew
tcp.options.echo
tcp.options.echoreply
tcp.options.eol
tcp.options.experimental
tcp.options.md5
tcp.options.mss
tcp.options.nop
tcp.options.qs
tcp.options.rvbd.probe
tcp.options.rvbd.trpy
tcp.options.sack
tcp.options.sack_perm
tcp.options.scps
No. Destination Protocol Length Info
:f4:ca Broadcast ARP 56 Who has 10.250.12.63? Tell 10.250.24.132
:f4:ca Broadcast ARP 56 Who has 10.250.14.188? Tell 10.250.24.132
gFug_c4:07:... Broadcast ARP 56 Who has 10.250.7.255? Tell 10.250.16.137
eTec_1a:aa:... Broadcast ARP 56 Who has 10.250.19.72? Tell 10.250.14.76
gFug_ce:85:... Broadcast ARP 56 Who has 10.250.24.45? Tell 10.250.23.229
7.113 224.0.0.251 MDNS 110 Standard query 0x0000 AAAA Android_DCDKY6P2.local, "QM" question A Android_DCDKY6P2.local, "QM" question
5.158 224.0.0.251 MDNS 180 Standard query 0x0000 ANY nearby-presence-nsd-67cfb6bb-4f56-483a-be58-6a5ef1ddb982._nearbypresence._tcp.local, "QM" question SRV
56:88ff:feb... ff02::fb MDNS 200 Standard query 0x0000 ANY nearby-presence-nsd-67cfb6bb-4f56-483a-be58-6a5ef1ddb982._nearbypresence._tcp.local, "QM" question SRV
acka_52:c1:... Broadcast ARP 60 Who has 10.250.13.221? Tell 10.250.1.1
acka_52:c1:... Broadcast ARP 60 Who has 10.250.19.52? Tell 10.250.1.1
acka_52:c1:... Broadcast ARP 60 Who has 10.250.18.239? Tell 10.250.1.1
eTec_7c:6a:... Broadcast ARP 56 Who has 10.250.14.0? Tell 10.250.22.163
gFug_77:f4:... Broadcast ARP 56 Who has 10.250.16.58? Tell 10.250.12.169
chno_4c:a9:... Broadcast ARP 56 Who has 10.250.12.45? Tell 10.250.15.229
chno_4c:a9:... Broadcast ARP 56 Who has 10.250.24.94? Tell 10.250.15.229
work_d9:30:... Broadcast ARP 56 Who has 10.250.30.247? Tell 10.250.24.24
acka_52:c1:... Broadcast ARP 60 Who has 10.250.22.217? Tell 10.250.1.1
:55:82 Broadcast ARP 56 Who has 10.250.15.129? Tell 10.250.15.162
3.184 224.0.0.251 MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(13)._spotify-connect._tcp.local SRV 0 0 53175 DESKTOP-04EMKPM.local
14345 23.271682800 fe80::37f0:c9b0:bb6... ff02::fb MDNS 355 Standard query response 0x0000 PTR Spotify Desktop Launcher(13)._spotify-connect._tcp.local SRV 0 0 53175 DESKTOP-04EMKPM.local
14346 23.271842800 Intel_b7:65:1a Broadcast ARP 56 Who has 10.250.2.21? Tell 10.250.23.184
14347 23.271947300 Intel_b7:65:1a Broadcast ARP 56 Who has 10.250.3.17? Tell 10.250.23.184
Frame 1: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{1F9
Ethernet II, Src: ChongqingFug_5c:5d:91 (5c:ba:ef:5c:5d:91), Dst: IPv6mcast_01 (33:33:00:00:00:01)
Internet Protocol Version 6, Src: fe80::e69b:298f:fe60:2e24, Dst: ff02::1
Internet Control Message Protocol v6
Transmission Control Protocol: Protocol
Packets: 14347 · Dropped: 0 (0.0%)
Profile: Default

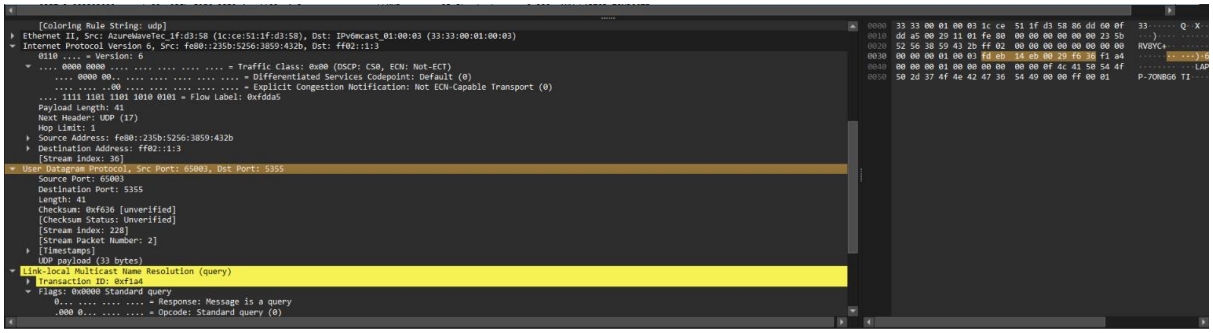

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
110 0.139126700 10.250.18.98 104.26.11.31 TCP 55 54793 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
111 0.152212300 8.8.4.4 10.250.18.98 TCP 66 443 → 55619 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
152 0.220657800 104.26.11.31 10.250.18.98 TCP 66 443 → 54793 [ACK] Seq=1 Ack=2 Win=16 Len=0 SLE=1 SRE=2
184 0.360979900 10.250.18.98 148.113.17.93 TCP 55 64534 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
185 0.371835600 148.113.17.93 10.250.18.98 TCP 60 443 → 64534 [ACK] Seq=0 Ack=2 Win=501 Len=0
186 0.371889300 10.250.18.98 148.113.17.93 TCP 54 [TCP Dup ACK 184#1] [TCP ACKed unseen segment] 64534 → 443 [ACK] Seq=2 Ack=1 Win=254 Len=0
340 0.529022200 148.113.17.93 10.250.18.98 TCP 66 [TCP Previous segment not captured] 443 → 64534 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
402 0.641664200 104.46.162.229 10.250.18.98 TCP 60 443 → 58818 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
403 0.672110800 10.250.18.98 192.179.20.84 TCP 55 49269 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
513 0.839119100 192.179.20.84 10.250.18.98 TCP 66 443 → 49269 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
514 0.894247700 10.250.18.98 192.179.20.84 TCP 55 50576 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
559 0.985867500 192.179.20.84 10.250.18.98 TCP 66 443 → 50576 [ACK] Seq=1 Ack=2 Win=1021 Len=0 SLE=1 SRE=2
672 1.246564200 10.250.11.3 10.250.18.98 TLSv1.2 168 Application Data
674 1.247179500 10.250.18.98 10.250.11.3 TLSv1.2 164 Application Data
776 1.448244500 10.250.18.98 149.154.170.200 TLSv1.2 178 Application Data
949 1.597874100 149.154.170.200 10.250.18.98 TLSv1.2 174 Application Data
950 1.620152500 10.250.18.98 10.250.11.3 TCP 164 [TCP Retransmission] 56384 → 8009 [PSH, ACK] Seq=1 Ack=115 Win=253 Len=110
1010 1.652951300 10.250.18.98 149.154.170.200 TCP 54 61481 → 443 [ACK] Seq=125 Ack=121 Win=254 Len=0
1044 1.698502900 10.250.11.3 10.250.18.98 TCP 66 8009 → 56384 [ACK] Seq=115 Ack=111 Win=1515 Len=0 SLE=1 SRE=111
1055 1.731433200 10.250.18.98 10.250.15.228 MS-DO 58 KeepAlive Message
1149 1.777640500 13.232.16.242 10.250.18.98 TLSv1.2 2550 Application Data, Application Data
1150 1.777701800 10.250.18.98 13.232.16.242 TCP 54 50946 → 443 [ACK] Seq=1 Ack=2497 Win=255 Len=0
Frame 110: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{1
Ethernet II, Src: AzureWaveTec_ed:bf:ea (a8:e2:91:ed:bf:ea), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:
Internet Protocol Version 4, Src: 10.250.18.98, Dst: 104.26.11.31
Transmission Control Protocol, Src Port: 54793, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
0000 4c d5 87 52 c1 00 a8 e2 91 ed bf ea 08 00 45 00
0010 00 29 9b 87 40 00 80 06 00 00 0a fa 12 62 68 1a
0020 0b 1f d6 09 01 bb 00 6d 2f ab 07 98 34 98 50 10
0030 00 ff 90 b0 00 00 00

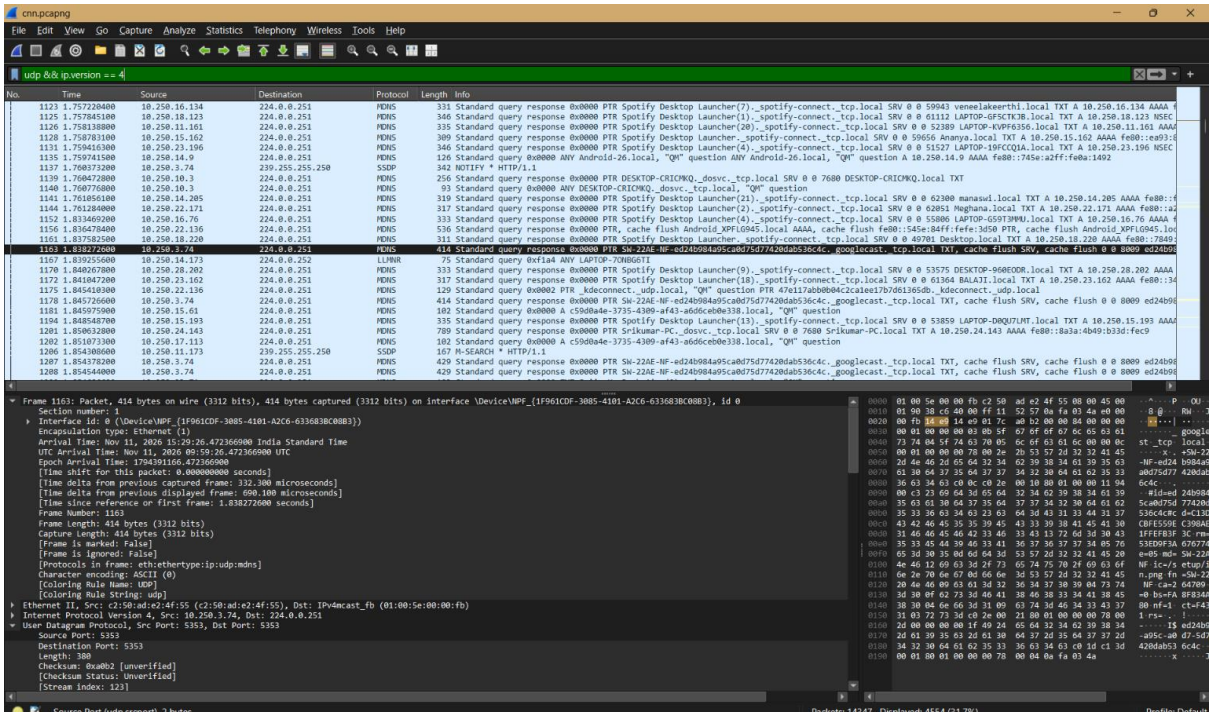

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
84 0.095248500 10.250.9.77 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
85 0.095316000 10.250.0.167 224.0.0.251 MDNS 472 Standard query response 0x0000 PTR Sujith's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 5328
86 0.095495300 fe80::76:4724:7c0c:... ff02::fb MDNS 492 Standard query response 0x0000 PTR Sujith's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 5328
88 0.095963100 fe80::c6a6:94d9:63c... ff02::1:3 LLMNR 88 Standard query 0xd9d2 ANY Gayathri
89 0.096029500 10.250.22.133 224.0.0.252 LLMNR 68 Standard query 0xd9d2 ANY Gayathri
93 0.097122600 10.250.16.53 224.0.0.251 MDNS 457 Standard query response 0x0000 PTR roop's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 55918
94 0.097297600 fe80::1c2f:5cae:e44... ff02::fb MDNS 477 Standard query response 0x0000 PTR roop's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 55918
97 0.097464700 10.250.30.22 224.0.0.251 MDNS 425 Standard query response 0x0000 PTR Mothe's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 49326
98 0.098186500 fe80::c64:4ae:380:e... ff02::fb MDNS 445 Standard query response 0x0000 PTR Mothe's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 49326
103 0.098896400 10.250.13.25 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SRV 0 0 53654 Harshika.local TXT A 10.
107 0.099209300 10.250.30.144 224.0.0.251 MDNS 327 Standard query response 0x0000 PTR Spotify Desktop Launcher(12)._spotify-connect._tcp.local SRV 0 0 57792 ShanmukhiPC.local TXT A
108 0.099349400 fe80::81bc:cd73:d20... ff02::fb MDNS 347 Standard query response 0x0000 PTR Spotify Desktop Launcher(12)._spotify-connect._tcp.local SRV 0 0 57792 ShanmukhiPC.local TXT A
112 0.187661300 fe80::e69b:298f:fe6... ff02::1:3 LLMNR 95 Standard query 0x2c8a ANY DESKTOP-VB4LQTD
113 0.187804200 10.250.22.225 224.0.0.252 LLMNR 75 Standard query 0x2c8a ANY DESKTOP-VB4LQTD
116 0.188731600 10.250.22.171 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(2)._spotify-connect._tcp.local SRV 0 0 62051 Meghana.local TXT A 10.2
117 0.189035000 fe80::a25f:9a18:7fb... ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher(2)._spotify-connect._tcp.local SRV 0 0 62051 Meghana.local TXT A 10.2
132 0.194692500 10.250.8.96 224.0.0.251 MDNS 1081 Standard query response 0x0000 TXT, cache flush PTR _airplay._tcp.local PTR Dinesh's MacBook Air._airplay._tcp.local TXT, cache
133 0.194853100 fe80::40b:f854:4eac... ff02::fb MDNS 1101 Standard query response 0x0000 TXT, cache flush PTR _airplay._tcp.local PTR Dinesh's MacBook Air._airplay._tcp.local TXT, cache
135 0.195003400 fe80::2524:79d4:6d1... ff02::1:3 LLMNR 84 Standard query 0xe73e A wpad
136 0.195038100 10.250.17.162 224.0.0.252 LLMNR 64 Standard query 0xe73e A wpad
137 0.195442500 fe80::2524:79d4:6d1... ff02::1:3 LLMNR 84 Standard query 0x9440 AAAA wpad
138 0.195468500 10.250.17.162 224.0.0.252 LLMNR 64 Standard query 0x9440 AAAA wpad
Frame 108: Packet, 347 bytes on wire (2776 bits), 347 bytes captured (2776 bits) on interface \Device\NF
Ethernet II, Src: CloudNetwork_e9:90:8f (3c:0a:f3:e9:90:8f), Dst: IPv6mcast_fb (33:33:00:00:00:fb)
Internet Protocol Version 6, Src: fe80::81bc:cd73:d20:f0ac, Dst: ff02::fb
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp && ip
Packet list Display filter Enter a display filter ... Find Cancel
Options: Narrow & Wide Case sensitive Backwards Multiple occurrences
No. Time Source Destination Protocol Length Info
93 0.097122600 10.250.16.53 224.0.0.251 MDNS 457 Standard query response 0x0000 PTR roop's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 55918
97 0.097464700 10.250.30.22 224.0.0.251 MDNS 425 Standard query response 0x0000 PTR Mothe's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 49326
103 0.098896400 10.250.13.25 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher(6)._spotify-connect._tcp.local SRV 0 0 53654 Harshika.local TXT A 10.
107 0.099209300 10.250.30.144 224.0.0.251 MDNS 327 Standard query response 0x0000 PTR Spotify Desktop Launcher(12)._spotify-connect._tcp.local SRV 0 0 57792 ShanmukhiPC.local TXT A
113 0.187804200 10.250.22.225 224.0.0.252 LLMNR 75 Standard query 0x2c8a ANY DESKTOP-VB4LQTD
116 0.188731600 10.250.22.171 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(2)._spotify-connect._tcp.local SRV 0 0 62051 Meghana.local TXT A 10.2
132 0.194692500 10.250.8.96 224.0.0.251 MDNS 1081 Standard query response 0x0000 TXT, cache flush PTR _airplay._tcp.local PTR Dinesh's MacBook Air._airplay._tcp.local TXT, cache
136 0.195038100 10.250.17.162 224.0.0.252 LLMNR 64 Standard query 0xe73e A wpad
138 0.195468500 10.250.17.162 224.0.0.252 LLMNR 64 Standard query 0x9440 AAAA wpad
140 0.197382900 10.250.17.162 224.0.0.251 MDNS 70 Standard query 0x0000 A wpad.local, "QM" question
142 0.197732800 10.250.17.162 224.0.0.251 MDNS 70 Standard query 0x0000 AAAA wpad.local, "QU" question
145 0.198525900 10.250.16.82 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
151 0.201519100 10.250.18.123 224.0.0.251 MDNS 346 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 61112 LAPTOP-GF5CTKJB.local T
163 0.306814700 10.250.24.193 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
165 0.310849500 10.250.15.99 224.0.0.251 MDNS 171 Standard query response 0x0000 A, cache flush 0.0.0.0 NSEC, cache flush f7f9d24e-2fda-49be-982b-245e65f66e07.local
166 0.312981400 10.250.14.173 224.0.0.252 LLMNR 75 Standard query 0xcdcf ANY LAPTOP-7ONBG6TI
173 0.320345100 10.250.16.26 224.0.0.251 MDNS 186 Standard query response 0x0000 AAAA, cache flush :: NSEC, cache flush 04335293-60c2-44ab-9e0a-5a2cf678d51c.local
174 0.320552300 10.250.10.3 224.0.0.251 MDNS 77 Standard query 0x0000 PTR _dosvc._tcp.local, "QU" question
176 0.320817300 10.250.18.98 10.250.10.3 MDNS 299 Standard query response 0x0000 PTR LAPTOP-BN8992KC._dosvc._tcp.local SRV 0 0 7680 LAPTOP-BN8992KC.local TXT A 10.250.18.98 AAAA
233 0.414875200 10.250.0.7 224.0.0.251 MDNS 100 Standard query 0x0000 PTR _companion-link._tcp.local, "QU" question PTR _rdlink._tcp.local, "QU" question
Frame 136: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface \Device\NPF_{1F961CDF-3085-4101-A2C6-633683BC08B3}, id 0
Ethernet II, Src: AzureWaveTec_ef:34:ba (60:ff:9e:ef:34:ba), Dst: IPv4mcast_fc (01:00:5e:00:00:fc)
Internet Protocol Version 4, Src: 10.250.17.162, Dst: 224.0.0.252
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 50
Identification: 0xe644 (58948)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0xd5de [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.17.162
Destination Address: 224.0.0.252
[Stream index: 44]
User Datagram Protocol, Src Port: 58542, Dst Port: 5355
Link-local Multicast Name Resolution (query)
Transaction ID: 0xe73e
Flags: 0x0000 Standard query
Questions: 1
Answer RRs: 0


cnn.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp.port == 5355
No. Time Source Destination Protocol Length Info
61 0.088715800 fe80::8a90:9e3e:ccc… ff02::1:3 LLMNR 89 Standard query 0xfc4b ANY Harshitha
70 0.091326500 10.250.18.16 224.0.0.252 LLMNR 69 Standard query 0xfc4b ANY Harshitha
88 0.095963100 fe80::c6a6:94d9:63c… ff02::1:3 LLMNR 88 Standard query 0xd9d2 ANY Gayathri
89 0.096029500 10.250.22.133 224.0.0.252 LLMNR 68 Standard query 0xd9d2 ANY Gayathri
112 0.187661300 fe80::e69b:298f:fe6… ff02::1:3 LLMNR 95 Standard query 0x2c8a ANY DESKTOP-VB4LQTD
113 0.187804200 10.250.22.225 224.0.0.252 LLMNR 75 Standard query 0x2c8a ANY DESKTOP-VB4LQTD
135 0.195003400 fe80::2524:79d4:6d1… ff02::1:3 LLMNR 84 Standard query 0xe73e A wpad
136 0.195038100 10.250.17.162 224.0.0.252 LLMNR 64 Standard query 0xe73e A wpad
137 0.195442500 fe80::2524:79d4:6d1… ff02::1:3 LLMNR 84 Standard query 0x9440 AAAA wpad
138 0.195468500 10.250.17.162 224.0.0.252 LLMNR 64 Standard query 0x9440 AAAA wpad
164 0.307401900 fe80::235b:5256:385… ff02::1:3 LLMNR 95 Standard query 0xcdcf ANY LAPTOP-7ONBG6TI
166 0.312981400 10.250.14.173 224.0.0.252 LLMNR 75 Standard query 0xcdcf ANY LAPTOP-7ONBG6TI
304 0.497920700 fe80::8a90:9e3e:ccc… ff02::1:3 LLMNR 89 Standard query 0x984f ANY Harshitha
305 0.498011800 10.250.18.16 224.0.0.252 LLMNR 69 Standard query 0x984f ANY Harshitha
430 0.712960700 fe80::235b:5256:385… ff02::1:3 LLMNR 95 Standard query 0xcdcf ANY LAPTOP-7ONBG6TI
431 0.713036100 10.250.14.173 224.0.0.252 LLMNR 75 Standard query 0xcdcf ANY LAPTOP-7ONBG6TI
530 0.917967100 fe80::8a90:9e3e:ccc… ff02::1:3 LLMNR 89 Standard query 0x984f ANY Harshitha
531 0.918838600 10.250.18.16 224.0.0.252 LLMNR 69 Standard query 0x984f ANY Harshitha
835 1.475453500 fe80::235b:5256:385… ff02::1:3 LLMNR 95 Standard query 0xf1a4 ANY LAPTOP-7ONBG6TI
836 1.475511700 10.250.14.173 224.0.0.252 LLMNR 75 Standard query 0xf1a4 ANY LAPTOP-7ONBG6TI
1166 1.839137000 fe80::235b:5256:385… ff02::1:3 LLMNR 95 Standard query 0xf1a4 ANY LAPTOP-7ONBG6TI
1167 1.839255600 10.250.14.173 224.0.0.252 LLMNR 75 Standard query 0xf1a4 ANY LAPTOP-7ONBG6TI
Frame 1166: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface \Device\NPF_{1F961CDF-3085-4101-A2C6-633683BC08B3}, id 0
Section number: 1
Interface id: 0 (\Device\NPF_{1F961CDF-3085-4101-A2C6-633683BC08B3})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 15:29:26.473231300 India Standard Time
UTC Arrival Time: Nov 11, 2026 09:59:26.473231300 UTC
Epoch Arrival Time: 1794391166.473231300
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 25.700 microseconds]
[Time delta from previous displayed frame: 363.625300 milliseconds]
[Time since reference or first frame: 1.839137000 seconds]
Frame Number: 1166
Frame Length: 95 bytes (760 bits)
Capture Length: 95 bytes (760 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:udp:llmnr]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: AzureWaveTec_1f:d3:58 (1c:ce:51:1f:d3:58), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
Destination: IPv6mcast_01:00:03 (33:33:00:01:00:03)
.... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)
0000 33 33 00 01 00 03 1c ce 51 1f d3 58 86 dd 60 0f
0010 dd a5 00 29 11 01 fe 80 00 00 00 00 00 00 23 5b
0020 52 56 38 59 43 2b ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 01 00 03 fd eb 14 eb 00 29 f6 36 f1 a4
0040 00 00 00 01 00 00 00 00 00 00 0f 4c 41 50 54 4f
0050 50 2d 37 4f 4e 42 47 36 54 49 00 00 ff 00 01


[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:udp:llmnr]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: AzureWaveTec_1f:d3:58 (1c:ce:51:1f:d3:58), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
Internet Protocol Version 6, Src: fe80::235b:5256:3859:432b, Dst: ff02::1:3
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
.... 1111 1101 1101 1010 0101 = Flow Label: 0xfdda5
Payload Length: 41
Next Header: UDP (17)
Hop Limit: 1
Source Address: fe80::235b:5256:3859:432b
Destination Address: ff02::1:3
[Stream index: 36]
User Datagram Protocol, Src Port: 65003, Dst Port: 5355
Link-local Multicast Name Resolution (query)
Transaction ID: 0xf1a4
Flags: 0x0000 Standard query
0... .... = Response: Message is a query
.000 0... .... = Opcode: Standard query (0)
.... .0.. .... = Conflict: None
Internet Protocol Version 6 (ipv6), 40 bytes
Packets: 14347 · Displayed: 1


[Coloring Rule String: udp]
Ethernet II, Src: AzureWaveTec_1f:d3:58 (1c:ce:51:1f:d3:58), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
Internet Protocol Version 6, Src: fe80::235b:5256:3859:432b, Dst: ff02::1:3
0110 .... = Version: 6
.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
.... 0000 00.. .... = Differentiated Services Codepoint: Default (0)
....00 .... = Explicit Congestion Notification: Not ECN-Capable Transport (0)
.... 1111 1101 1101 1010 0101 = Flow Label: 0xfdda5
Payload Length: 41
Next Header: UDP (17)
Hop Limit: 1
Source Address: fe80::235b:5256:3859:432b
Destination Address: ff02::1:3
[Stream index: 36]
User Datagram Protocol, Src Port: 65003, Dst Port: 5355
Source Port: 65003
Destination Port: 5355
Length: 41
Checksum: 0xf636 [unverified]
[Checksum Status: Unverified]
[Stream index: 228]
[Stream Packet Number: 2]
[Timestamps]
UDP payload (33 bytes)
Link-local Multicast Name Resolution (query)
Transaction ID: 0xf1a4
Flags: 0x0000 Standard query
0... .... = Response: Message is a query
.000 0... .... = Opcode: Standard query (0)

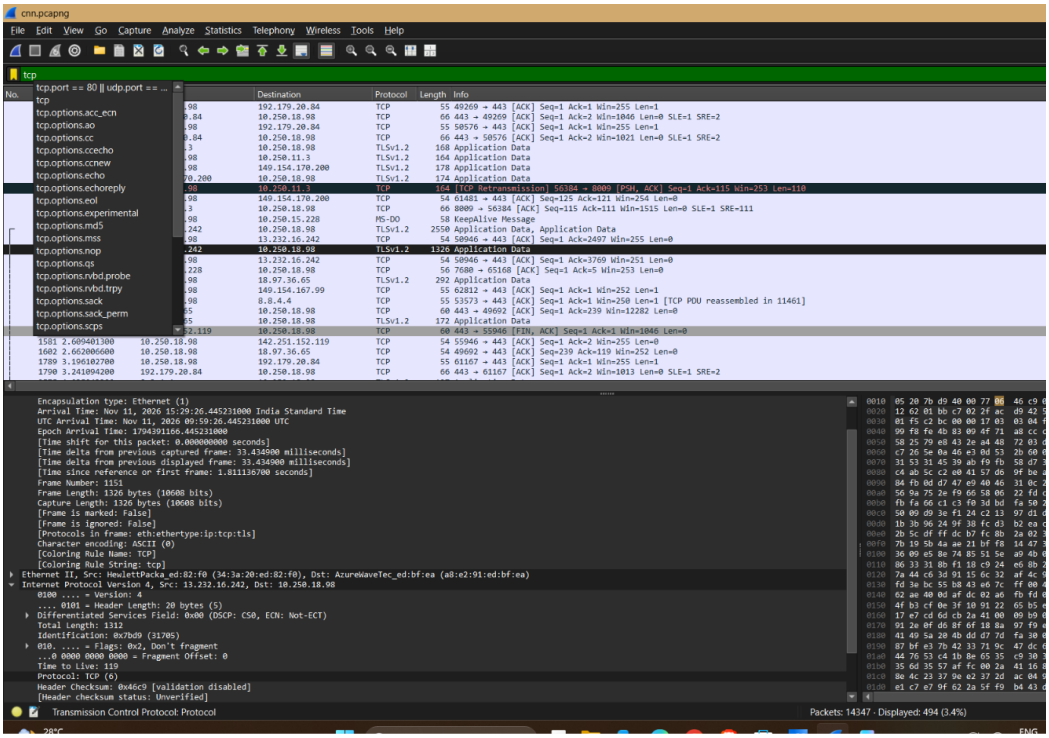


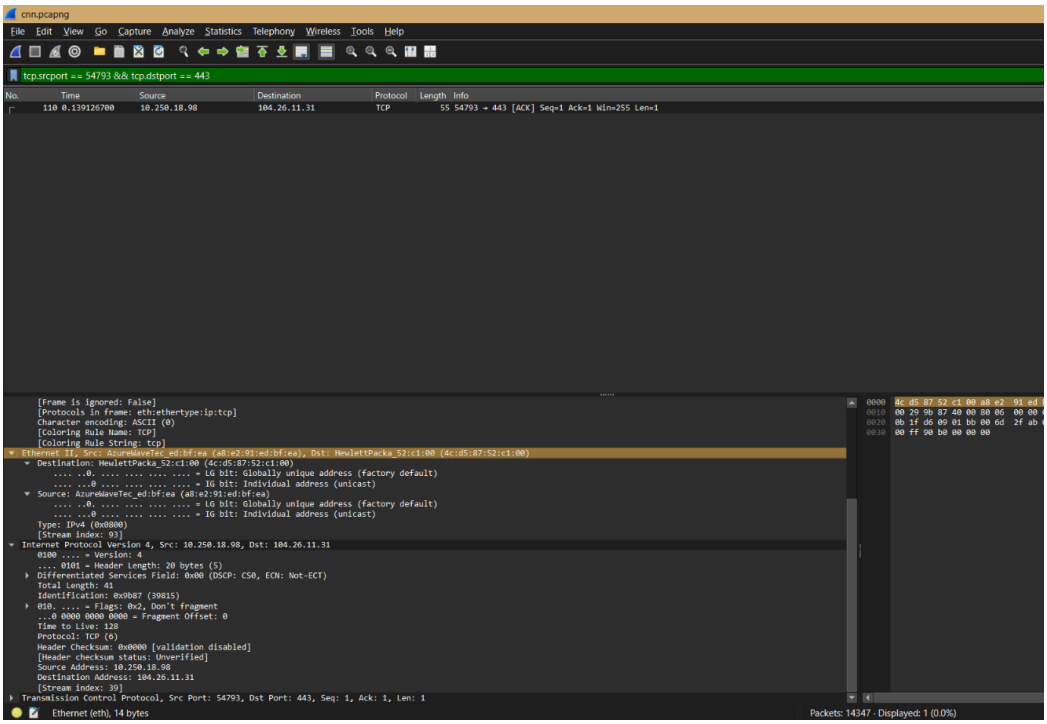
UDP packet – Final observations

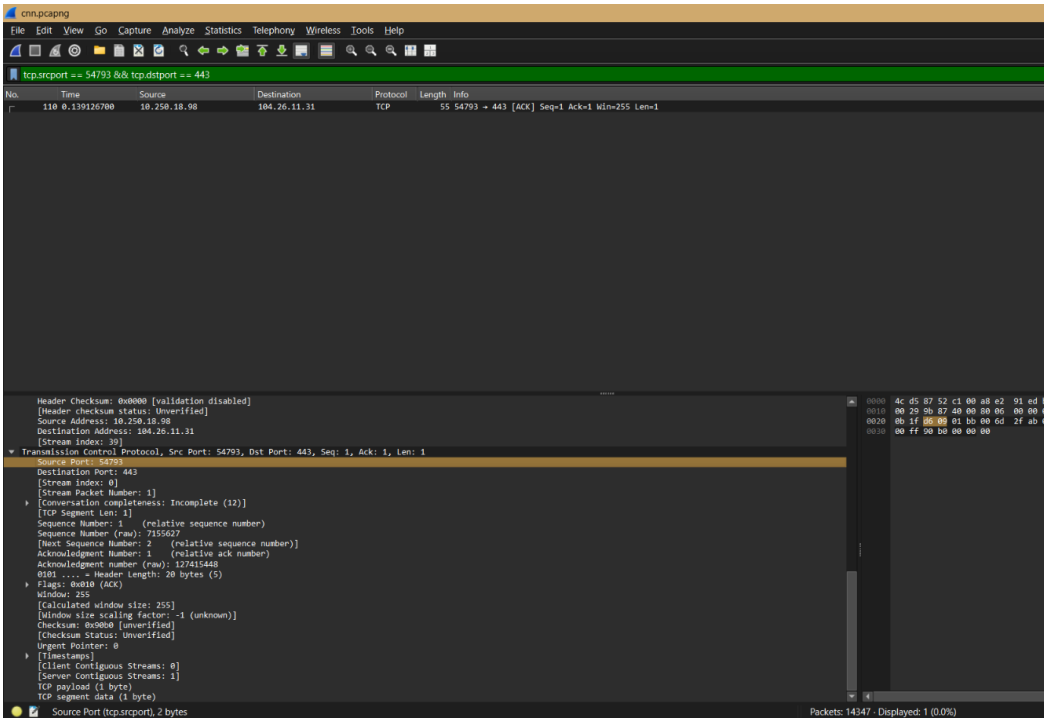
Parameter	Your actual value
Frame Length	414 bytes
Ethernet Header Length	14 bytes
IP Header Length	20 bytes
IP Total Length	400 bytes
UDP Header Length	8 bytes
UDP Length	380 bytes
UDP Payload Length	372 bytes
Source IP	10.250.3.74
Destination IP	224.0.0.251
Source Port	5353
Destination Port	5353

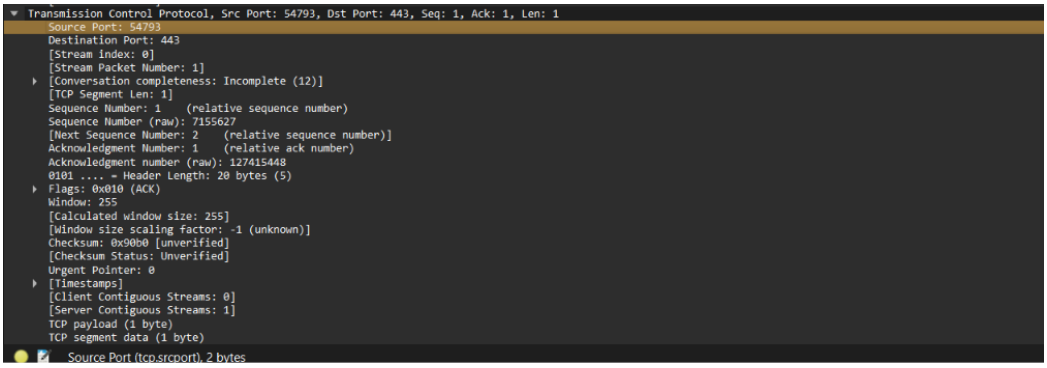
Protocol UDP

TCP









TCP packet – Final verification

Parameter	Verified value
Source Port	54793
Destination Port	443
TCP Header Length	20 bytes
TCP Segment Data Length	1 byte
TCP Flags	ACK
TCP Payload	1 byte

OBSERVATION

Parameter	TCP Packet	UDP Packet
Frame Length	55 bytes	414 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	41 bytes	400 bytes
Transport Segment Length	21 bytes	380 bytes
Payload Length	1 byte	372 bytes

Result

Successfully captured and analysed TCP and UDP packets using Wireshark.